

VENKATRAMAN R

Data Science & AI Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Entry-level Data Scientist and AI/ML Engineer with hands-on experience building 8 end-to-end machine learning, forecasting, and generative AI projects. Achieved 0.97 AUC-ROC on a mobile transaction fraud detection model (FraudGuard) using XGBoost with SMOTE and SHAP explainability. Skilled in Python, SQL, cloud platforms (AWS, Azure), and BI tools (Power BI), with practical experience in RAG pipelines, model deployment, and data pipeline automation. Seeking an entry-level Data Science or AI/ML role to apply these skills to real-world problems.

Technical Skills

Programming & Cloud: Python, SQL, AWS (S3, EC2), Azure, PostgreSQL

Generative AI & NLP: LLMs, LangChain (LCEL), RAG Pipelines, FAISS Vector Search, OpenAI API, Groq Llama-3.3-70B, Hugging Face, Prompt Engineering

Machine Learning: Scikit-Learn, XGBoost, Random Forest, SMOTE, Feature Engineering, Cross-Validation, Optuna, SHAP Explainability

Deep Learning & MLOps: TensorFlow, Keras, LSTM, CNN, RNN, Docker, GitHub Actions (CI/CD)

Data Analysis & BI: SQL Querying, EDA, Data Cleaning, Model Evaluation, Dashboard Development, A/B Testing, Hypothesis Testing, Pandas, NumPy, Power BI (DAX, Power Query, Star Schema), Streamlit, Plotly, Advanced Excel

Technical Projects

AskMyData – GenAI Data-Analysis Agent | *Python, LangChain, FAISS, Docker*

Jan 2026 – May 2026

- Built an autonomous AI data-analysis agent using LangChain LCEL that routes natural-language queries into automated SQL and Pandas workflows.
- Implemented FAISS-based semantic vector indexing for database metadata mapping, reducing hallucination risk and improving downstream code-generation accuracy by 18%.
- Engineered a secure, Dockerized Python execution sandbox with input sanitization, multi-turn conversational memory, and automated error-handling loops.
- Integrated Plotly visualizations, delivering an end-to-end pipeline from natural-language query to secure dashboard output; deployed live on Streamlit Cloud.

FraudGuard – Mobile Transaction Fraud Detection | *Python, XGBoost, SMOTE, SHAP*

Jan 2026 – May 2026

- Trained an XGBoost classifier on a mobile transaction fraud dataset, achieving 0.97 AUC-ROC and 0.92 F1-score.
- Corrected severe class imbalance using SMOTE oversampling and tuned hyperparameters with Optuna to improve generalization.
- Applied SHAP explainability to interpret model decisions and support transparent, auditable fraud-flagging logic.
- Deployed the model as a live Streamlit web app for real-time transaction risk scoring.

SalesPulse Forecaster – Deep Learning Sales Forecasting | *Python, TensorFlow, LSTM, AWS*

Sep 2025 – Dec 2025

- Built a two-layer LSTM network (128 and 64 units) with TensorFlow/Keras to generate recursive 12-month sales forecasts with dynamic confidence bands (MAE \$126K, MAPE 27.66%).
- Architected a data ingestion pipeline backed by PostgreSQL and hosted on AWS to support reliable upload-to-export workflows.
- Evaluated forecast accuracy using MAE and MAPE, iterating on network depth and unit sizing to reduce error.
- Documented the full modeling workflow in a reproducible Jupyter notebook for stakeholder review.

SalesPulse Analytics – BI Dashboard & ETL Pipeline | *Power BI, DAX, Python, Excel*

May 2025 – Aug 2025

- Authored 44 custom DAX measures over a star-schema Power BI model spanning \$12.6M+ of global sales data to compute metrics like customer LTV and YoY margin variance.
- Built a modular ETL pipeline processing 51,000+ relational records into clean Parquet and Excel outputs using Pandas and openpyxl.
- Developed an interactive Excel-based maintenance-tracking dashboard mapping KPI variables across 50,000 vehicle fleet service records.

Education

Bachelor of Engineering (B.E.) – Automobile Engineering, VISTAS, Chennai

2021 – 2024

CGPA: 7.25

Certifications

- Professional Graduate Certification in Data Science & Artificial Intelligence – Boston Institute of Analytics (June 2026)